truly equal? Are all humans equal to one another biologically? Let us try to translate the most famous line of the American Declaration of Independence into biological terms:

> We hold these truths to be self-evident, that all men are **created equal**, that they are **endowed** by their **Creator** with certain **unalienable rights**, that among these are life, **liberty**, and the pursuit of **happiness**.

According to the science of biology, people were not ‘created’. They have evolved. And they certainly did not evolve to be ‘equal’. The idea of equality is inextricably intertwined with the idea of creation. The Americans got the idea of equality from Christianity, which argues that every person has a divinely created soul, and that all souls are equal before God. However, if we do not believe in the Christian myths about God, creation and souls, what does it mean that all people are ‘equal’? Evolution is based on difference, not on equality. Every person carries a somewhat different genetic code, and is exposed from birth to different environmental influences. This leads to the development of different qualities that carry with them different chances of survival. ‘Created equal’ should therefore be translated into ‘evolved differently’.

Just as people were never created, neither, according to the science of biology, is there a ‘Creator’ who ‘endows’ them with anything. There is only a blind evolutionary process, devoid of any purpose, leading to the birth of individuals. ‘Endowed by their Creator’ should be translated simply into ‘born’.

Similarly, there are no such things as rights in biology. There are only organs, abilities and characteristics. Birds fly not because they have a right to fly, but because they have wings. And it’s not true that these organs, abilities and characteristics are ‘unalienable’. Many of them undergo constant mutations, and may well be completely lost over time. The ostrich is a bird that lost its ability to fly. So ‘unalienable rights’ should be translated into ‘mutable characteristics’.

And what are the characteristics that evolved in humans? ‘Life’, certainly. But ‘liberty’? There is no such thing in biology. Just like equality, rights and limited liability companies, liberty too is a political ideal rather than a biological phenomenon. From a purely biological viewpoint, there is little difference between the citizens of a republic and the subjects of a king. And what about ‘happiness’? So far biological research has failed to come